Cambridge IGCSE Geography 0460

Paper 4 – Alternative to Coursework
Gredos Fieldtrip Revision Pack
Santo António International School | Year 11 | 2026

Depth measurement using ranging rod

Measuring river width with tape

Sediment sampling in channel

Recording velocity data

This revision pack prepares you for Cambridge IGCSE Geography Paper 4 (Alternative to Coursework). It mirrors the structure of a real Paper 4 question paper, using data and fieldwork evidence collected during the **Gredos Fieldtrip, Portugal/Spain**. Three enquiries are covered: **Rivers**, **Ecosystems**, and **Settlement**.

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Section 1: Rivers Investigation

During the Gredos fieldtrip, students conducted a rivers investigation to test the hypothesis: *'River channel characteristics change downstream.'* Data was collected at three sites along a stream in the Gredos mountains, measuring depth, width, velocity, and sediment size.

Students using waders and ranging rods to measure channel depth

Stretching a tape across the channel to record river width

Sediment sampling — recording angularity and size (Powers Scale)

Preparing equipment at the upstream study site

1.1 Key Terminology

Cross-sectional area	Width × mean depth of the river channel (m^2)
Discharge	Volume of water flowing past a point per second (m^3/s) = CSA × velocity
Hydraulic radius	Cross-sectional area ÷ wetted perimeter — efficiency measure
Bedload	Coarse sediment (pebbles, gravel) rolled along the riverbed
Attrition	Particles knock together, becoming smaller and more rounded downstream
Hjulström curve	Shows velocity needed to erode, transport, or deposit sediment of different sizes
Bradshaw Model	Predicts how channel characteristics change downstream

1.2 Data Collection Methods

Depth: Ranging rod placed vertically at equal intervals across the channel width; depth recorded in cm.

Width: Tape measure stretched bank-to-bank at the waterline; recorded in metres.

Velocity: Flow meter / dog biscuit timed over 10 m; average of 3 repeats taken.

Sediment: 10 pebbles collected at random, b-axis measured, angularity rated on Powers Scale (1–6).

Wetted Perimeter: Tape measure follows channel bed shape; used to calculate hydraulic radius.

1.3 Fieldwork Data — Three Sites

Variable	Site 1 (Upstream)	Site 2 (Mid)	Site 3 (Downstream)
Width (m)	1.8	3.4	5.2
Mean depth (m)	0.12	0.19	0.27
CSA (m^2)	0.22	0.65	1.40
Velocity (m/s)	0.31	0.48	0.62
Discharge (m^3/s)	0.07	0.31	0.87

Mean sediment (mm)	48	32	19
Mean roundness (1–6)	2.1	3.4	4.6

1.4 Exam Practice Questions

[2 marks] State two methods used to measure river velocity in the field.

[4 marks] Using the data in Table 1.3, describe how discharge changes from Site 1 to Site 3.

[7 marks] Explain why sediment becomes smaller and more rounded further downstream. Use evidence from the fieldtrip data in your answer.

PEEL Answer Framework

Point — State your argument clearly

Evidence — Name data / place / statistic

Explain — Link evidence to the point (because / therefore)

Link — Connect back to the question

✓ Mark Scheme — 7 marks — Rivers Sediment Question

- P — Sediment becomes smaller and more rounded downstream (award 1 mark for clear point)
- E — Data: mean sediment decreases from 48 mm (Site 1) to 19 mm (Site 3); roundness increases from 2.1 to 4.6
- E — Attrition: particles collide during transport, breaking into smaller, rounder pieces
- E — Abrasion: sediment scrapes channel bed/banks, losing sharp edges over distance
- L — Therefore, Bradshaw Model prediction of decreasing particle size downstream is supported
- QWC: Logical sequence of argument; geographical terminology used throughout (max 2 extra marks)

Section 2: Ecosystems Investigation

Students investigated vegetation distribution across two contrasting habitats in the Gredos region: a **pine forest floor** and a **meadow grassland**. The hypothesis tested was: *'Species diversity is greater in the meadow than in the pine forest understory.'*

Identifying species within a 0.5 m² quadrat frame

Students recording percentage cover using gridded quadrat

Group briefing before random sampling in pine forest

Close inspection of ground-level vegetation in quadrat

2.1 Sampling Methods

Random sampling was used to avoid bias. A 10 m × 10 m study area was marked out using ranging rods and tape measures. Random co-ordinates were generated by rolling dice and the 0.5 m² gridded quadrat was placed at each intersection. For each quadrat, students recorded:

- Species present (identified using field guide)
- Percentage cover of each species (estimated using quadrat grid)
- Percentage bare ground
- Height of tallest plant (cm)
- Light intensity (lux meter reading)

2.2 Key Terminology

Quadrat	A square frame (0.25 m² or 0.5 m²) placed on the ground to sample vegetation
Random sampling	Using random numbers to choose sample locations — reduces bias
Percentage cover	Proportion of the quadrat area covered by each plant species (%)
Species diversity	The variety and abundance of different species in an area
Abiotic factor	Non-living environmental factor (light, moisture, pH) affecting plant growth
Succession	Gradual change in species composition of an ecosystem over time
Simpson's Index	Diversity index: higher value = greater diversity (max = 1)

2.3 Results — Mean % Cover by Habitat

Species	Pine Forest (% cover)	Meadow (% cover)
Grasses	8	54
Heather/shrubs	12	18
Mosses/lichens	31	6

Wildflowers	2	22
Bare ground/litter	47	0
Total species recorded	6	14
Simpson's Index (D)	0.58	0.87

2.4 Exam Practice Questions

[2 marks] State two advantages of using random sampling when conducting a quadrat survey.

[4 marks] Using Table 2.3, compare vegetation cover in the pine forest and meadow habitats.

[7 marks] Explain why species diversity is lower in the pine forest understory than in the meadow. Use evidence from your fieldwork data and geographical knowledge.

PEEL Answer Framework

Point — State your argument clearly

Evidence — Name data / place / statistic

Explain — Link evidence to the point (because / therefore)

Link — Connect back to the question

✓ Mark Scheme — 7 marks — Ecosystems Diversity Question

- P — Pine forest has lower species diversity than meadow (D = 0.58 vs 0.87)
- E — 47% bare ground/litter in forest vs 0% in meadow; only 6 species in forest vs 14
- E — Pine canopy creates deep shade (low light intensity) — limits photosynthesis of ground plants
- E — Needle litter creates acidic, nutrient-poor soil — restricts species that can survive
- E — Meadow receives full sunlight, has higher soil moisture and pH variety — supports more niches
- L — Therefore, abiotic differences (light, soil pH) explain the biodiversity gradient
- QWC: Reference to abiotic factors; use of data to support claims; logical argument structure

Section 3: Settlement Investigation — Environmental Quality Survey

Students conducted an Environmental Quality Assessment (EQA) in a traditional village in the Gredos region, testing the hypothesis: *'Environmental quality declines from the village centre towards the periphery.'* A bi-polar evaluation method was used, rating 8 indicators on a scale of –2 to +2.

Students conducting EQA survey along a village street

Recording observations on EQA survey sheets

3.1 Bi-Polar EQA Indicators

Building condition	Very poor –2 ... +2 Very good
Road/pavement quality	Very poor –2 ... +2 Very good
Green space / planting	None –2 ... +2 Abundant
Noise level	Very loud –2 ... +2 Very quiet
Litter / cleanliness	Very dirty –2 ... +2 Very clean
Traffic congestion	Severe –2 ... +2 Minimal
Pedestrian safety	Very unsafe –2 ... +2 Very safe
Natural environment	Degraded –2 ... +2 High quality

3.2 EQA Results by Zone

EQA Indicator	Zone A (Centre)	Zone B (Transition)	Zone C (Periphery)
Building condition	+2	+1	–1
Road/pavement quality	+1	+1	–2
Green space	0	+1	+2
Noise	+1	0	+2
Litter/cleanliness	+2	+1	–1
Traffic	–1	0	+2
Pedestrian safety	+1	+1	0
Natural environment	0	+1	+2

Total score (/16)	**+6**	**+6**	**+4**

3.3 Exam Practice Questions

[2 marks] Describe the bi-polar evaluation method used to assess environmental quality.

[4 marks] Using Table 3.2, describe how environmental quality varies between Zone A and Zone C.

[7 marks] To what extent does the data support the hypothesis that environmental quality declines from the village centre to the periphery? Use evidence in your answer.

PEEL Answer Framework

Point — State your argument clearly

Evidence — Name data / place / statistic

Explain — Link evidence to the point (because / therefore)

Link — Connect back to the question

✓ Mark Scheme — 7 marks — Settlement EQA Question

- P — Data partially supports the hypothesis: overall score drops from +6 to +4 at periphery
- E — Building condition falls from +2 (centre) to –1 (periphery); road quality from +1 to –2
- E — However, green space, noise, and natural environment IMPROVE at periphery (+2 each)
- E — Traffic is better at periphery (+2) — suggests less congestion away from centre
- L — Hypothesis is only partially supported; quality varies by indicator, not uniformly
- AO3: Recognises limitations of method (subjectivity of bi-polar scoring; small sample)
- QWC: Uses data; structured argument; geographical terminology (bi-polar, EQA, zones)

Section 4: Exam Technique & Paper 4 Guidance

Paper 4 lasts **1 hour 30 minutes** and is worth **60 marks**. All questions relate to **your own fieldwork** — there is no pre-release resource booklet. You must recall methods, data, and analysis from your investigation.

4.1 Command Word Guide

State / Identify	**1–2 marks**	Give a fact. No explanation needed. One word or phrase.
Describe	**2–4 marks**	Say what you see in data — use figures, quote the data, use comparators.
Explain	**4–7 marks**	Give reasons — use 'because', 'therefore', 'this leads to'.
Suggest	**2–4 marks**	Give a reasoned possibility — you won't know for certain.
Evaluate	**6–7 marks**	Weigh up — consider strengths AND limitations, reach a conclusion.
Compare	**3–4 marks**	Must refer to BOTH items being compared in each point.

4.2 Avoiding Common Mistakes

- Not quoting specific data — always include figures from your data table.
- Writing only one side of an argument for evaluate/to what extent questions.
- Confusing 'describe' and 'explain' — describe = what, explain = why.
- Forgetting to link back to the hypothesis or question in the conclusion.
- Mixing up method limitations with general problems — be specific to YOUR fieldwork.
- Writing in bullet points — extended questions require continuous prose.

4.3 Time Management

Marks available	Suggested time	What to do
1–2 marks	2–3 mins	Short, direct answers — do not over-write
3–4 marks	5–7 mins	2–3 developed points with data
6–7 marks	12–15 mins	PEEL structure; 3–4 paragraphs; conclusion

4.4 Fieldtrip Photo Summary

Rivers — depth & width measurement in the channel

Ecosystems — quadrat sampling in the field

Settlement — EQA survey in the village

4.5 Final Checklist

- ■ I can name the location and date of my fieldtrip (Gredos region, Spain/Portugal)
- ■ I can state the hypothesis for each of the three investigations
- ■ I can describe at least TWO data collection methods per investigation
- ■ I can quote specific data from tables to support my answers
- ■ I can explain the limitations of my methods (e.g. subjectivity, sample size)
- ■ I can evaluate whether the results support or refute the hypothesis
- ■ I can write an extended answer using the PEEL framework
- ■ I know the command words and how many points each mark requires

✓ Mark Scheme — General Mark Scheme Principles — Paper 4

- Level 1 (1–2): Basic, vague points; little or no data reference; no geographical terminology
- Level 2 (3–4): Developed points; some data; mostly accurate terminology
- Level 3 (5–6): Well-structured argument; accurate data; good terminology; some evaluation
- Level 4 (7): Full PEEL structure; data throughout; both sides evaluated; clear conclusion
- QWC marks: Awarded for quality of written communication on extended questions